pole, which is devoid of photo receptors; hence this region is called **blind spot.**

Mechanism of Vision

When light enters the eyes, it gets refracted by the **cornea, aqueous humor and lens and it is focused** on the retina and excites the rod and cone cells. The photo pigment consists of **Opsin**, the protein part and **Retinal**, a derivative of vitamin A. Light induces dissociation of retinal from opsin and causes the structural changes in opsin. This generates an action potential in the photoreceptor cells and is transmitted by the optic nerves to the visual cortex of the brain, via bipolar cells, ganglia and optic nerves, for the perception of vision.

Refractive errors of eye

Myopia (near sightedness): The affected person can see the nearby objects but not the distant objects. This condition may result due to an elongated eyeball or thickened lens; so that the image of distant object is formed in front of the yellow spot. This error can be corrected using concave lens that diverge the entering light rays and focuses it on the retina.

Hypermetropia (long sightedness): the affected person can see only the distant objects clearly but not the objects nearby. This condition results due to a shortened eyeball and thin lens; so the image of closest object is converged behind the retina. This defect can be overcome by using convex lens that converge the entering light rays on the retina.

Presbyopia: Due to aging, the lens loses elasticity and the power of accommodation. Convex lenses are used to correct this defect.

Astigmatism is due to the rough (irregular) curvature of cornea or lens. Cylindrical glasses are used to correct this error (Figure 10.16).

Cataract: Due to the changes in nature of protein, the lens becomes opaque. It can be corrected by surgical procedures.

Visual pigments for colour vision are i) the red cones having the visual pigment, Erythropsin is sensitive to long wavelength close to 560 nm. ii) The green cones having the pigment, chloropsin is sensitive to medium wavelength of 530 nm iii) the blue cones having the pigment, cyanopsin is sensitive to short wavelength of 420 nm.

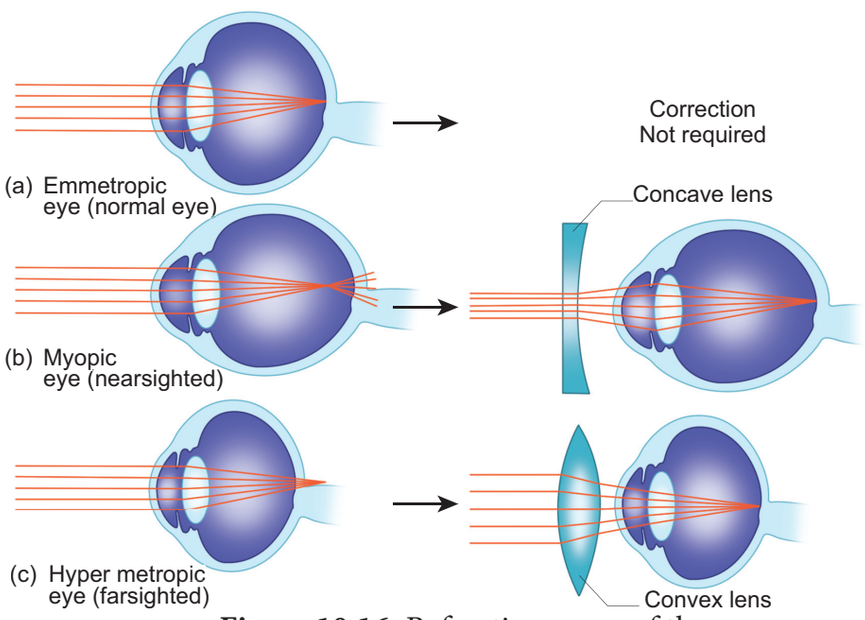


Figure 10.16 Refractive errors of the eye

XI Std Zoology Chapter-10 EM.indd 192 30/12/2022 13:55:39